Build with AI

H2S

Solution Challenge

Team Name: Syntax Terrors
Team Leader Name: Tanish Sanghvi
Problem Statement Title: [Smart Resource Allocation] Open Innovation
The Challenge & Impact: Educational institutions are paralyzed by "Data Stagnation", where critical academic records are locked in fragmented silos, forcing students, staffs and institutions into hundreds of hours of manual and error-prone data entry.

Solution to Problem

The ‘Department Ledger Portal’ an AI-first Master Ledger" system that leverages Google Gemini 2.5 Flash to transform fragmented, static academic records into structured and actionable institutional intelligence.

The Concept of “Master Ledger” Intelligence

AI-first ecosystem designed to transform "**Static Documents**" into “Structured Institutional Intelligence”

Multimodal parsing of physical marksheets and certificates into digital records

How It Solves the Problem

20 Minutes
Manual Entry

→

10 Second AIParse
90% Manual Effort Reduction

Bridging Data Silos

Actionable Database

Placement Readiness

Real-Time Oversight - Unlocks trapped metrics

The X-Factor (Core Innovation)

Smart Analysis

Gemini 2.5 Flash

Field-Specific Extraction

(Name, GPA, etc.)

High Accuracy

Career Pulse

Student

SWOT Analysis

Strengths	Weaknesses
Opportunities	Treats

Career Roadmap

UN SDG - Alignment / Impact

SDG 4 — Quality Education
Eliminates data barriers to track and improve student academic outcomes

SDG 8 — Decent Work & Economic Growth
Career Pulse accelerates placement readiness and student employability

SDG 9 — Industry, Innovation & Infrastructure
AI-first infrastructure replacing manual academic system

Opportunities & Comparison

A high-impact shift from manual record-keeping to a Zero-Trust AI Document-to-Ledger Protocol, reclaiming hundreds of administrative hours through high-speed multimodal intelligence.

Problem Resolution Strategy

1. Ultra-Fast Processing - 10-second AI parsing replaces 20-minute manual entry
2. High Data Accuracy - Minimizes human errors in document-to-ledger conversion
3. Structured Intelligence - Converts static records into searchable, high-value data
4. 90% Effort Reduction - Up to 90% manual entry effort reduction

USP & Impact

1. Unique Selling Point - AI-powered document-to-ledger system with secured logging
2. Career Readiness - Provides SWOT analysis and personalized roadmaps
3. Institutional Oversight - Enables real-time, data-driven insights for faculty

System Comparison	Existing System (Google Forms)	Proposed System (Department Ledger Portal)
Data Entry Speed	Manual entry (~20 min/student)	AI parsing (~10 sec/doc)
Intelligence	Static responses (No analysis)	AI-assisted insights with validation layers
Accuracy	High risk of human error	AI validation layers
Data Structure	Isolated, flat responses	Unified Master Ledger
Data Integrity	No duplicate prevention	Context-aware detection
Searchability	Limited spreadsheet filtering	Advanced dashboards & search
Student Value	No career guidance	Career Pulse: SWOT & Roadmaps
Governance	Basic/No access control	Secure RBAC & Audit Trails

Note :- Comparison reflects current MVP capabilities vs typical spreadsheet workflow.

List of Features Offered

The Department Ledger Portal offers a suite of AI-driven features, integrating multimodal document parsing with institutional governance to transform raw records into a secure and searchable digital ecosystem.

AI Intelligence Core

Smart Analysis - Gemini 2.5 Flash parses marksheets and certificates into ledger fields with accuracy.

Career Pulse - Automated profiling engine generates placement readiness scores and career roadmaps.

Context-Aware Processing - AI cross-references existing data to prevent duplicate records during auto-fill.

Governance & Security

Multi-Role RBAC - Secure, tiered access control for Admin, Faculty, and Student roles.

Immutable Audit Trail - Append-only logging of every administrative action ensures transparency and integrity.

Secure Role-based Approval - All new registrants require manual Admin elevation to access the dashboard.

Academic & Management

6-Section Master Ledger - Modules for Academics, Achievements, Activities, Placements, Projects, and Skills.

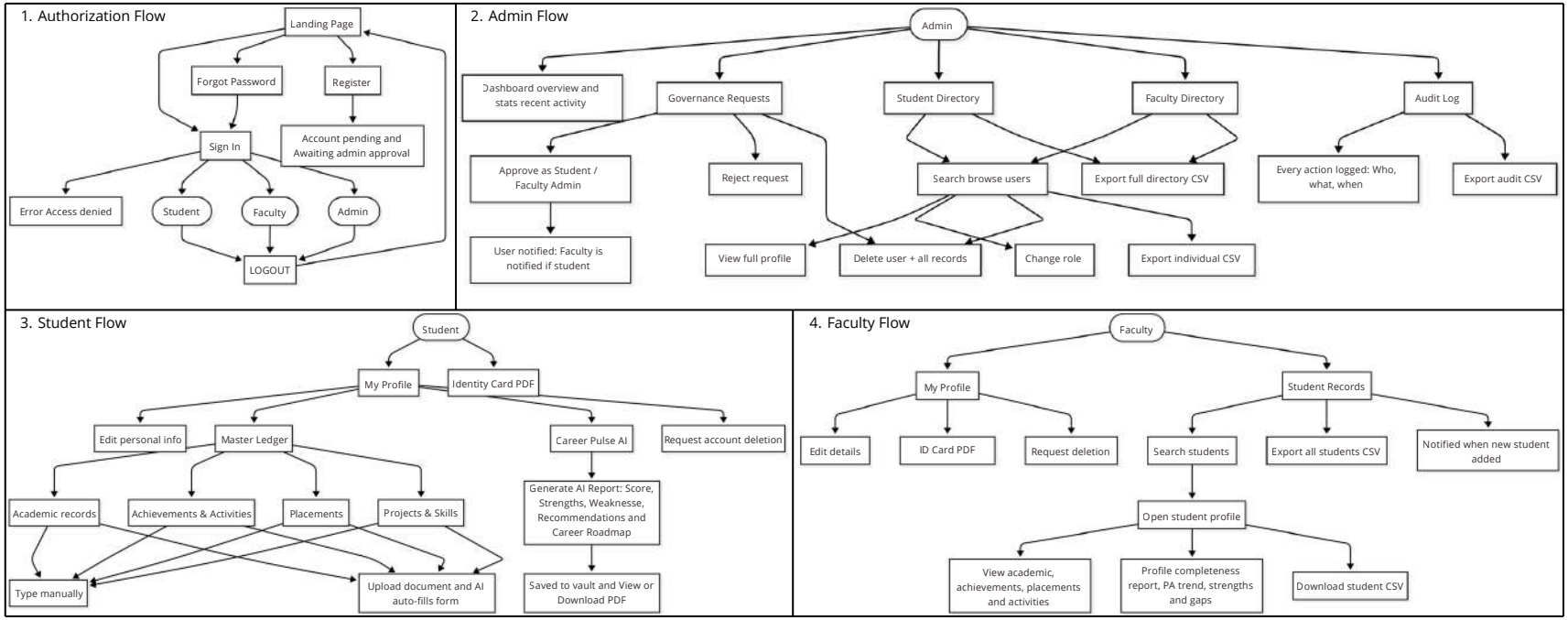
Faculty Oversight - Real-time departmental dashboards featuring advanced search, filtering, and high-density performance metrics.

Data Portability - One-click professional CSV exports for anonymized departmental reporting and analysis.

Digital Identity Cards - Dynamically generated, high-resolution printable ID cards for all authorized users.

Flow diagrams of the proposed solution

The Department Ledger Portal features an integrated, multi-role operational flow that orchestrates secure onboarding, administrative governance and faculty oversight, while empowering students with AI-driven career intelligence and a unified master ledger.



Technologies Used

The Department Ledger Portal is built on a high-performance, serverless architecture that combines Next.js 16+ and Vercel for delivery with the intelligence of Gemini AI and the secured real-time data management of Firebase.

CORE INTELLIGENCE (GOOGLE AI)

GEMINI 2.5 FLASH
Multimodular Parsing that extracts structured data from images & PDFs (marksheet, certificates)

CAPABILITIES
Document parsing, auto form filling, placement readiness analysis

FRONTEND & USER EXPERIENCE

NEXT.JS 16+ (PAGES ROUTER)
Core application framework (high performance React)

TAILWIND CSS
Responsive UI with modern glassmorphic design

BACKEND & CLOUD INFRASTRUCTURE (VERCEL)

VERCEL CLOUD
Global hosting with fast deployment

SERVERLESS API ROUTES
Low-latency backend logic on edge network

DATABASE & SECURITY (FIREBASE / GOOGLE)

FIRESTORE
Scalable NoSQL database in real-time and secured row level data protection

FIREBASE AUTHENTICATION
Secure login & role-based access

Estimated Implementation Cost & ROI

Asset-Light Infrastructure model with a serverless architecture that scales with usage, minimizing costs while maximizing administrative time savings.

OVERALL COST STRUCTURE

Low-Cost & Scalable Architecture

→ Infrastructure (Vercel & Firebase) of Pay-as-you-go model, fits within free-tier for initial pilots

AI Intelligence (Gemini API)

→ ~$0.01 per document.parse, eliminates manual data entry costs

Deployment Model

→ Serverless SaaS with zero upfront hardware or maintenance

ESTIMATED ROI (PILOT ASSUMPTIONS)

High Impact, Measurable Gains

→ 90% Effort Reduction Removes manual data entry workload

Estimated Reclaimable Time

→ 200+ hours of Annual time savings per department

High Data Accuracy

→ Eliminates human errors, ensures reliable digital records

Note :- Based on: 20 min manual entry per student × 600 students/year/department = 200 hrs. AI parsing at ~10 sec/doc = ~1.7 hrs. Reduction = 99.2%, conservatively stated as 90%.

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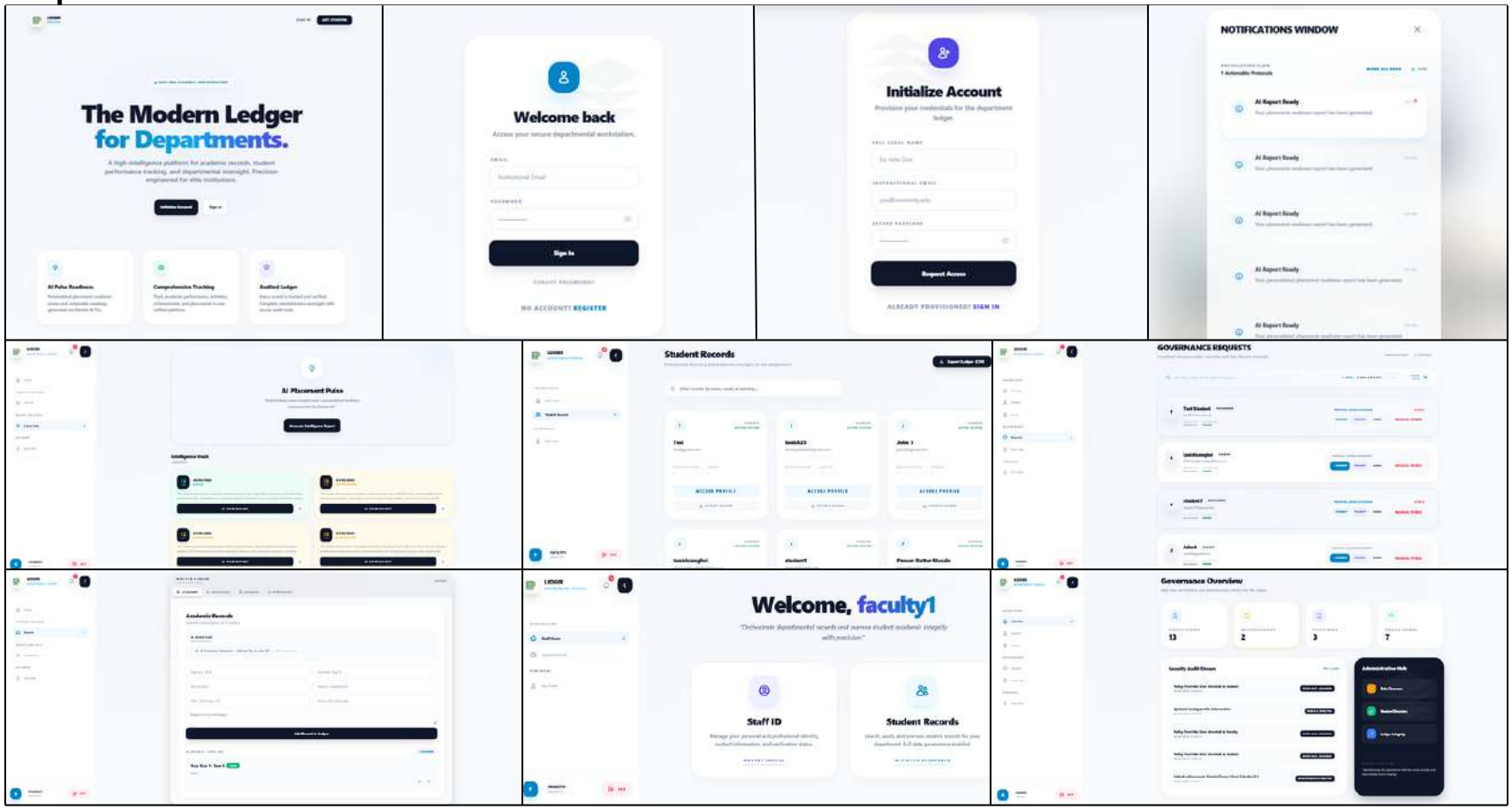
Snapshots of MVP



Additional Details & Future Roadmap

The Department Ledger Portal is built as a foundational layer for institutional intelligence, evolving from a working departmental tool into a universal platform with AI-Automation, cryptography and longitudinal career tracking.

PHASE 1
CURRENT MVP

- **AI Document Parsing** (PDF/Image → ledger fields)
- **Career Pulse** (SWOT + Readiness Score)
- **3-Role RBAC System** (Student / Faculty / Admin)
- **Immutable Audit Log** (Append Only Chain)
- **Real-time Notifications** (Notifications via System)

[Built & Employed]

PHASE 2
3 – 6 MONTHS

- **Bulk Import Service** (Batch PDF processing)
- **Verified Transcript Engine** (AI-based official transcripts)
- **Alumni Ledger Extension** (Longitudinal career tracking)
- **AI Peer Mentorship** (Skill-based student matching)
- **Predictive Placement AI** (Probability score per student)

[In Development]

PHASE 3
6 – 12 MONTHS

- **Multi-Institution Support** (One portal, multiple colleges)
- **Cryptographically Signed PDFs** (Exports via Crypto API)
- **Edge AI Deployment** (Local inference in campuses)
- **Offline Mode** (PWA with local caching)
- **University-wide Analytics** (Placement + GPA dashboards)

[Planned]

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Project Links

The following links provide full access to the 'Department Ledger Portal' codebase, live environment and functional demonstrations.

- **GitHub Public Repository:**
  https://github.com/tanish-jain-225/Department-Ledger-Portal
- **Working Prototype (MVP):**
  https://department-ledger-portal.vercel.app
- **Video Demo (3 Minutes):**
  https://drive.google.com/drive/folders/1a2Bklmxa7QwUcPPeY83r4hymXAEa6kbX?usp=sharing

"TRANSFORMING STATIC DOCUMENTS INTO STRUCTURED, GOVERNED INTELLIGENCE - ONE UPLOAD AT A TIME"

KEY HIGHLIGHTS

- 2 AI FEATURES
- 3 ROLE SYSTEMS
- 12 FIRESTORE COLLECTIONS

BUILD TOOLS

- GOOGLE GEMINI
- FIREBASE
- NEXT.JS
- VERCEL

SYNTAX TERRORS

TEAM: SYNTAX TERRORS
LEADER: TANISH SANGHVI